

1) What is Simple linear regression? State how to use Simple linear regression with its relevant formula. (5 marks)

→ Simple linear regression is a statistical method used to understand the relationship between two variables

- i) One dependent Variable (the outcome or response)
- ii) One independent Variable (predictor or explanatory Variable)

In Simple, linear it predicts the value of dependent Variable based on the value of independent Variable

Formula for Simple linear regression

General formula for Simple linear regression is

$$Y = mx + c$$

where Y is the dependent variable (the variable we want to predict)

x is the independent Variable (the variable we use to make prediction)

c is the intercept (the value of Y when $x=0$)

m is the slope of regression line
(change in Y for one unit of change in x)

How to use Simple linear regression

① Collect Data: Gather data for both dependent and independent Variable. For example, if you want to predict house prices (Y) based on area (sq.foot), collect historical data on both variables

② Plot The data: Create a scatter plot to visualize the relationship between independent Variable (x) and dependent variable (Y)

3) Calculate the regression line

• The slope (m) can be calculated using formula

$$m = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

• The intercept c can be calculated as using formula

$$Y = mx + c$$

$$\therefore c = \bar{y} - m\bar{x}$$

where $\bar{x}$ and $\bar{y}$ are means of the X and Y variables

4) plot the regression line

Draw the regression line using the equation

$Y = mx + c$. This line represents best fit for the data points, minimizing the sum of squared differences between the observed and predicted values

5) Make predictions: Use the regression line equation to make predictions for dependent variable (Y) given new values of the independent variable (X) for example, if you have equation

$$Y = 1.30 +$$

$$Y = 0.04x + 1.30$$

(from petrol price example, you can predict the petrol price for any given demand level (x))

Consider ~~price~~ predicting the price of house based on Square foot

Suppose your data shows that for every additional 1000 sq ft price increases by \$50,000

Your regression equation might be

$$\text{price} = 100,000 + 200 \times \text{Square foot}$$

Here \$100,000 is the ~~base~~ intercept (base price) and

\$200 per square foot is the slope

If you want to predict the price of house with 2,500 square feet

$$\text{price} = 100,000 + 200 \times 2500$$

$$= 600,000$$

$$\text{predicted price} = 600,000$$

2) What is multiple linear regression? State how to use multiple linear regression with its relevant formula.

→ Multiple linear regression is a statistical technique used to model relationship between one dependent variable and two or more independent variables. Unlike simple linear regression which deals with only one independent variable, multiple linear regression allows us to analyze how several factors simultaneously influence the dependent variable.

This method is particularly useful when you want to predict an outcome based on the multiple predictors. It provides a way to quantify effect of each predictor while controlling for the influence of the other.

Formula for multiple linear regression

General formula for multiple linear regression is

$$Y = a + b_1x_1 + b_2x_2 + \dots + b_nx_n$$

where

- Y is the dependent variable (outcome you are trying to predict)
- x_1, x_2 are independent variables (the predictors)
- a is the intercept (the predicted value of Y when all x are zero)
- $b_1, b_2 \dots b_n$ are the coefficients (slopes) that represents the change in Y for a one-unit change in each corresponding x .

How to use multiple linear regression

1. Collect and prepare data

- Gather the data for dependent Variable and all relevant independent variables

For example: If you want to predict house price, your dataset might include variables like square foot, number of bedrooms, location and age of the house

2. Check assumptions

- Linearity: The relationship between the dependent Variable and each independent Variable
- Independence: The observations should be independent of each other
- Homoscedasticity: The variance of errors should be consistent across all levels of the independent variables
- Normality: The residuals (errors) should be normally distributed

3. Fit the model

- Use statistical software to estimate the coefficients $b_1, b_2, \dots, b_n$ and intercept a
- The software will provide regression equation, which can be used to make predictions

4. Interpret the result

- The intercept a indicates the predicted value of Y when all independent variables are zero
- Each coefficient b_i shows how much dependent variable Y changes for one-unit increase in corresponding independent variable X_i , holding all other variables constant

5. Make predictions

- Use the regression equation to predict dependent Variable Y for new values of independent Variables $X_1, X_2, \dots, X_n$